

PARKINSON'S DISEASE

PT Assessment and PT Management

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PT Assessment

- **Observation**

- **Behaviour of patient**
- **Posture of the patient**
- **Patient's facial attitude**
- **Skin condition (any superficial lesion)**
- **Tremor**
- **Type and variation of tremor**

- **Examination**

- **Motor examination**
 1. **Tone**
 2. **Muscle tightness**
 3. **ROM**

PT Assessment

- **Examination**

- **Motor examination**

4. **Manual Muscle Testing**
5. **Test for Bradykinesia**
6. **Tremor – Location, Persistence and Severity**

- **Postural control and balance**

- **Done by “Berg Balance Scale”**

- **Gait Analysis**

- **All the gait parameters should be examined**

- **Coordination**

- **Finger Nose Finger Test**
 - **Finger to Finger test**

PT Assessment

- **Examination**
 - **Sensory examination**
 - **Superficial**
 1. **Pain**
 2. **Touch**
 3. **Temperature**
 - **Deep**
 1. **Pressure**
 2. **Vibration**
 3. **Joint Position Sense**
 - **Cortical**
 1. **Tactile localization**
 2. **Stereognosis**

PT Assessment

- **Examination**
 - **Cognitive Function examination**
 - **Memory, Orientation, Conceptual reasoning and judgement should be checked.**
 - **MMSE Scale can be used**
- **Chest and CVS Assessment**
 - **Breathing Pattern**
 - **Inspiratory-Expiratory Ratio**
 - **Respiratory & Heart Rate**
 - **Heart Sound**
 - **Respiratory Sound**

PT Assessment

- **Hoehn and Yahr Staging of Parkinson's Disease**

- **Stage 1**

- **Sign & Symptoms on only one side**
- **Mild Symptoms**
- **Symptoms inconvenient but not disabling**

- **Stage 2**

- **B/L Symptoms**
- **Minimal disability**
- **Balance normal**

- **Stage 3**

- **B/L Symptoms**
- **Mild postural imbalance**

PT Assessment

- **Hoehn and Yahr Staging of Parkinson's Disease**

- **Stage 4**

- **B/L involvement**
- **Postural instability**
- **Patient needs help**

- **Stage 5**

- **B/L involvement with more severity**
- **Severe Postural instability**
- **Patient restricted to bed/chair**

Aim of Physiotherapy

- **Improve mobility**
- **Improve muscle strength**
- **Improve flexibility**
- **Improve posture and balance**
- **Maintain a good breathing pattern**
- **Maintain and improve the level of function and independence**

PT Management

- **Relaxation**

- **Gentle rocking**
- **Rotational movement of limbs and trunk**
- **Stretching**
- **Positioning – Hook lying**
- **Diaphragmatic breathing**
- **Lifestyle modifications**

- **Flexibility**

- **Active assistive**
- **Passive movement**
- **PNF with resistance**
- **PNF Stretching – Contract relax (6 sec contraction followed by 10-30 sec assistive stretch)**

PT Management

- **Flexibility**

- **Unilateral bridging with trunk rotation**
- **Bilateral bridging with trunk rotation**
- **High kneeling with anterior pelvis translation**
- **MRP**
- **Sit to stand**
- **Weight bearing exercises**
- **Frenkel exercise**
- **Backward stepping**
- **Patient with PD experiences a high no of falling, so he/she should be taught how to get up after fall.**

PT Management

- **Balance Training**
 - **Weight shifting**
 - **Reaching**
 - **Axial rotation of head and trunk**
 - **Axial rotation with reaching**
 - **Sitting on inflated ball**
 - **Kneeling**
 - **Half kneeling**
 - **Standing**
 - **Overhead arm clapping**
 - **Heel raises**
 - **Toe Standing**

PT Management

- **Balance Training**

- **Toe walking**
- **Wall squats**
- **Chair raises**
- **Single limb stance with side kick and back kick**
- **Whole body vibration in Physioacoustic Chair**

- **Gait Training**

- **Straight**
- **Zig-Zag**
- **Side by Side**
- **Backward stepping**
- **Verbal command**
- **Functional training**

PT Management

- **Pulmonary rehab**

- **Diaphragmatic breathing**
- **Postural drainage**
- **Air shifting techniques**
- **Vibration and shaking**

- **Home Plan**

- **Make sure that the patient understood about the importance of self exercises**
- **Early morning warmups**
- **Self stretching and self resistive exercises**
- **Yoga**
- **Meditation**
- **Breathing Exercises**